NEET UG TARGET STANDARD | CLASS 12 CHEMISTRY • INORGANIC CHEMISTRY

UNIT: COORDINATION COMPOUNDS

Coordination Compounds

Complete Conceptual Tutorial, Theoretical Framework, and NEET-Aligned Assessment

1. Syllabus Context & NEET Exam Orientation

> **NEET Examination Profile:** In the NEET UG Chemistry paper, *Coordination Compounds* represents one of the highest-weightage chapters in Class 12 Inorganic Chemistry, routinely contributing **2 to 3 questions (8 to 12 marks)** every year. NEET questions in this chapter are renowned for testing precision in conceptual fundamentals: identifying primary vs. secondary valences, deducing IUPAC nomenclature, evaluating isomer counts (especially optical activity of chelated complexes), predicting magnetic moments using VBT and CFT, calculating crystal field stabilization energy (CFSE), and understanding synergic bonding in metal carbonyls.

Target Learning Objectives

Upon completing this tutorial, a competitive examination aspirant will be able to:

- Apply **Werner's coordination theory** to deduce the structural formulations, primary/secondary valences, precipitated moles of AgCl, and electrolytic solution conductivities of coordination complexes.
- Distinguish rigorously between **double salts** and **complex ions** based on their aqueous dissociation behavior.
- Identify and classify ligands according to charge, denticity (unidentate, didentate, hexadentate), chelation (and the resulting chelate effect), and ambidentate behavior.
- Determine central metal oxidation states and coordination numbers (counting σ-linkages exclusively).
- Apply current IUPAC nomenclature conventions to name mononuclear complexes and formulate chemical representations unambiguously.
- Classify and enumerate **structural isomers** (ionisation, solvate/hydrate, linkage, coordination) and **stereoisomers** (geometrical *cis*/*trans* and *fac*/*mer*; optical enantiomers).
- Predict hybridisation schemes, spatial geometries, spin states (inner vs. outer orbital), and spin-only magnetic moments using **Valence Bond Theory (VBT)**.
- Explain **Crystal Field Theory (CFT)** orbital splitting in octahedral (Δ_o) and tetrahedral (Δ_t) environments, apply the spectrochemical series, evaluate pairing energies (P), calculate CFSE, and rationalize $d-d$ transitions, complementary colors, and gemstone pigmentation.
- Describe **synergic bonding** in transition metal carbonyls ($\mathrm{M-C}$ σ-bond and $\mathrm{M \rightarrow CO}$ π-backbonding) and its structural/vibrational consequences.
- Identify high-yield analytical, metallurgical, biological, industrial, and medicinal applications of coordination compounds.

2. Alfred Werner's Theory of Coordination Compounds

Alfred Werner (1866–1919), recognized as the founder of modern coordination chemistry, explained the unique bonding and stoichiometry in transition metal addition compounds by proposing that metal ions exhibit two distinct types of valency:

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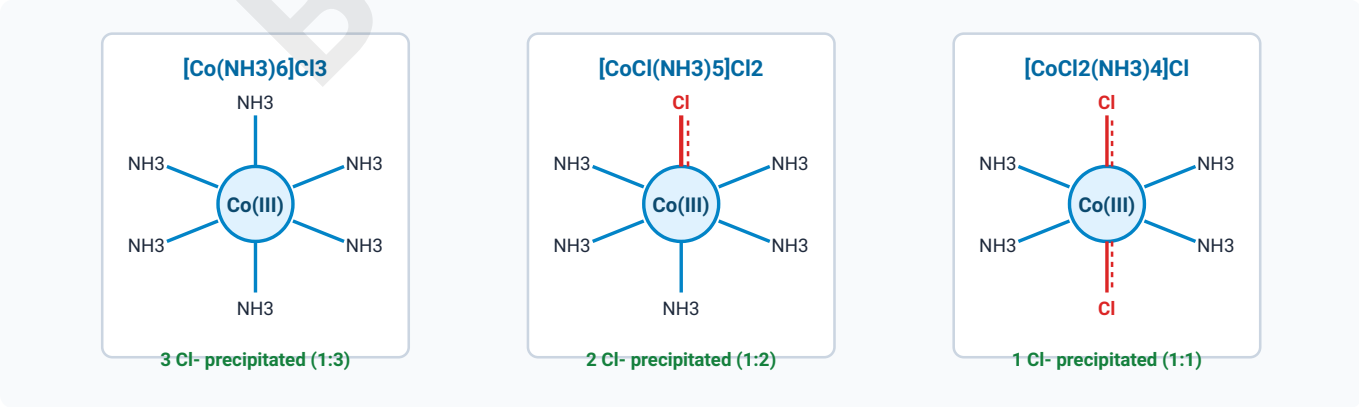
Postulates of Werner's Theory

1. **Two Types of Linkages:** In coordination compounds, central metals exhibit two types of linkages (valences): *primary valence* and *secondary valence*.
2. **Primary Valence:** Corresponds to the oxidation state of the metal ion. It is *ionisable*, non-directional in space, and is satisfied strictly by negative ions (anions).
3. **Secondary Valence:** Corresponds to the *coordination number* of the metal ion. It is *non-ionisable* and directional in space. It is satisfied by either neutral molecules (e.g., $\mathbf{NH_3}$, $\mathbf{H_2O}$) or negative ions (e.g., $\mathbf{Cl^-}$, $\mathbf{CN^-}$). The secondary valence is fixed for a given metal in a specific oxidation state (e.g., cobalt(III) universally exhibits a secondary valence of 6).
4. **Spatial Directionality (Coordination Polyhedra):** The groups bound by secondary linkages occupy definite, directional spatial positions around the central metal ion, generating characteristic geometries such as octahedral, tetrahedral, and square planar.

Experimental Validation: Cobalt(III) Chloride–Ammonia Complexes

Werner reacted cobalt(III) chloride with ammonia in varying proportions to obtain four distinct complexes. By treating 1 mole of each compound with excess cold aqueous silver nitrate ($\mathbf{AgNO_3}$) and measuring the electrolytic conductivity of the resulting solutions, he deduced their structural formulations:

Empirical Formula & Color	Modern Formulation	Moles of $\mathbf{AgCl}$ Precipitated (per mol)	Total Ions per Formula Unit	Electrolytic Conductance Type
$\mathbf{CoCl_3 \cdot 6NH_3}$ (Yellow)	$[Co(NH_3)_6]^{3+}3Cl^-$	$\mathbf{3\ mol\ AgCl}$	4 (1 complex cation + 3 chloride anions)	1:3 electrolyte
$\mathbf{CoCl_3 \cdot 5NH_3}$ (Purple)	$[CoCl(NH_3)_5]^{2+}2Cl^-$	$\mathbf{2\ mol\ AgCl}$	3 (1 complex cation + 2 chloride anions)	1:2 electrolyte
$\mathbf{CoCl_3 \cdot 4NH_3}$ (Green)	$[CoCl_2(NH_3)_4]^+Cl^-$	$\mathbf{1\ mol\ AgCl}$	2 (1 complex cation + 1 chloride anion)	1:1 electrolyte (*trans*-isomer)
$\mathbf{CoCl_3 \cdot 4NH_3}$ (Violet)	$[CoCl_2(NH_3)_4]^+Cl^-$	$\mathbf{1\ mol\ AgCl}$	2 (1 complex cation + 1 chloride anion)	1:1 electrolyte (*cis*-isomer)



Tutorial Figure 1: Structural Representation of Werner's Valencies in Cobalt(III) Ammine Complexes. Solid lines represent non-ionisable secondary valencies; combined solid-and-dashed lines denote chloride ions satisfying dual primary and secondary valencies.

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NEET High-Yield Distinction: Double Salts vs. Coordination Complexes

- **Double Salts:** Addition compounds formed by stoichiometric combinations of stable salts that dissociate *completely into individual constituent simple ions* when dissolved in water. They retain identity only in the solid state.
  • *Carnallite:* $\mathbf{KCl \cdot MgCl_2 \cdot 6H_2O \rightarrow K^+ + Mg^{2+} + 3Cl^- + 6H_2O}$
  • *Mohr's Salt:* $\mathbf{FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O \rightarrow Fe^{2+} + 2NH_4^+ + 2SO_4^{2-} + 6H_2O}$ (gives characteristic test for $\mathbf{Fe^{2+}}$)
  • *Potash Alum:* $\mathbf{KAl(SO_4)_2 \cdot 12H_2O \rightarrow K^+ + Al^{3+} + 2SO_4^{2-} + 12H_2O}$
- **Coordination Complexes:** Retain their molecular identity in both solid and aqueous states. The complex entity enclosed in square brackets does *not* dissociate into simple ions.
  • Potassium hexacyanidoferrate(II), $\mathbf{K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-}}$. The solution gives *no* test for free $\mathbf{Fe^{2+}}$ or $\mathbf{CN^-}$ ions.
  • Tetraamminecopper(II) sulphate, $\mathbf{[Cu(NH_3)_4]SO_4 \rightarrow [Cu(NH_3)_4]^{2+} + SO_4^{2-}}$, giving no test for free $\mathbf{Cu^{2+}}$.

3. Essential Definitions and Structural Terminology

(a) Coordination Entity & Central Atom/Ion

A **coordination entity** constitutes a central metal atom or cation bonded to a fixed number of ions or neutral molecules. For example, $\mathbf{[CoCl_3(NH_3)_3]}$ is a neutral coordination entity, while $\mathbf{[Fe(CN)_6]^{4-}}$ is an anionic entity, and $\mathbf{[Co(NH_3)_6]^{3+}}$ is a cationic entity.

The **central atom or ion** is the transition metal center to which a fixed number of donor groups are coordinated in a definite geometrical arrangement. Because it accepts electron pairs from ligands, the central atom/ion functions as a **Lewis acid**.

(b) Ligands and Their Classification

The ions or neutral molecules directly bonded to the central metal atom/ion in a coordination entity are called **ligands**. Because they donate lone pairs of electrons to form coordinate covalent σ-bonds with the central metal, ligands function as **Lewis bases**.

- **Unidentate (Monodentate) Ligands:** Ligands that bind to the metal center through a single donor atom. Examples: $\mathbf{Cl^-}$, $\mathbf{Br^-}$, $\mathbf{F^-}$, $\mathbf{H_2O}$ (donor: $\mathbf{O}$), $\mathbf{NH_3}$ (donor: $\mathbf{N}$), $\mathbf{OH^-}$, $\mathbf{CN^-}$, $\mathbf{CO}$.
- **Didentate (Bidentate) Ligands:** Ligands possessing two donor atoms that can coordinate simultaneously to a single metal center.
  • *Ethane-1,2-diamine* ($\mathbf{H_2NCH_2CH_2NH_2}$, abbreviated as 'en'): Neutral ligand with two donor nitrogen atoms.
  • *Oxalate ion* ($\mathbf{C_2O_4^{2-}}$, abbreviated as 'ox'): Symmetrical dinegative anion coordinating through two negatively charged oxygen atoms.
- **Polydentate Ligands:** Contain multiple donor atoms.
  • *Triaminotriethylamine* ($\mathbf{N(CH_2CH_2NH_2)_3}$): Branched tetradentate ligand coordinating through four nitrogen atoms.
  • *Ethylenediaminetetraacetate ion* ($\mathbf{EDTA^{4-}}$): An important **hexadentate** ligand that coordinates through two nitrogen atoms and four carboxylate oxygen atoms.
- **Chelating Ligands & The Chelate Effect:** When a di- or polydentate ligand binds a single metal ion through two or more donor atoms simultaneously, it forms a 5- or 6-membered heterocyclic ring known as a **chelate ring**. Complexes containing chelate rings are called *chelate complexes*.
  The Chelate Effect: Chelate complexes exhibit markedly greater thermodynamic stability than analogous complexes formed by monodentate ligands. This enhanced stability is predominantly an *entropy-driven phenomenon* ($\Delta S^\circ > 0$), as displacement of multiple monodentate molecules by a single polydentate ligand increases the net number of independent particles in solution:

$$[\mathrm{Ni(H_2O)_6}]^{2+} + 3\,\mathrm{en} \rightleftharpoons [\mathrm{Ni(en)_3}]^{2+} + 6\,\mathrm{H_2O} \quad (\Delta S^\circ > 0)$$

- **Ambidentate Ligands:** Monodentate ligands having two distinct potential donor atoms, either of which can coordinate to the central metal atom/ion:
  • *Nitrite ion* ($\mathrm{NO_2^-}$): Can bind through nitrogen ($\mathrm{M \leftarrow NO_2}$, named **nitrito-N** or nitro) or through oxygen ($\mathrm{M \leftarrow ONO}$, named **nitrito-O**).
  • *Thiocyanate ion* ($\mathrm{SCN^-}$): Can bind through sulfur ($\mathrm{M \leftarrow SCN}$, named **thiocyanato-S**) or through nitrogen ($\mathrm{M \leftarrow NCS}$, named **isothiocyanato** or thiocyanato-N).

(c) Coordination Number (CN)

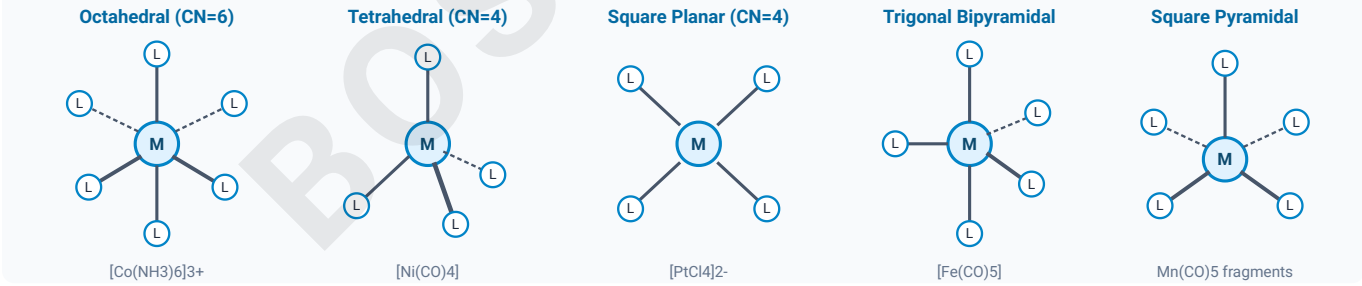
The **coordination number** of a central metal ion is defined as the *total number of ligand donor atoms directly bonded to the metal ion by σ-bonds*.

NEET Calculation Rule: $\mathrm{Coordination\ Number} = \sum(\mathrm{number\ of\ ligands} \times \mathrm{denticity})$.
• In $[\mathrm{PtCl_6}]^{2-}$, $\mathrm{CN} = 6 \times 1 = 6$.
• In $[\mathrm{Ni(NH_3)_4}]^{2+}$, $\mathrm{CN} = 4 \times 1 = 4$.
• In $[\mathrm{Fe(C_2O_4)_3}]^{3-}$, oxalate is didentate: $\mathrm{CN} = 3 \times 2 = 6$.
• In $[\mathrm{Co(en)_3}]^{3+}$, 'en' is didentate: $\mathrm{CN} = 3 \times 2 = 6$.
• In $[\mathrm{Ca(EDTA)}]^{2-}$, $\mathrm{EDTA^{4-}}$ is hexadentate: $\mathrm{CN} = 1 \times 6 = 6$.
Crucial note: π-bonds between metal and ligand are **never** counted toward the coordination number.

(d) Coordination Sphere, Counter Ions, & Coordination Polyhedra

The central metal atom/ion and its coordinated ligands are enclosed within square brackets, $[\ldots]$, collectively forming the non-ionisable **coordination sphere**. Ionisable species written outside the brackets are **counter ions** (e.g., in $\mathrm{K_4[Fe(CN)_6]}$, $[\mathrm{Fe(CN)_6}]^{4-}$ is the coordination sphere and $4\,\mathrm{K^+}$ are the counter ions).

The spatial arrangement of ligand donor atoms directly attached to the central metal defines the **coordination polyhedron** about the center:



Tutorial Figure 2: The Five Principal Coordination Polyhedra in Coordination Chemistry.

(e) Oxidation Number of the Central Atom

The **oxidation number** of the central atom in a complex is defined as the residual charge it would carry if all coordinated ligands were removed along with the shared electron pairs. It is denoted by a Roman numeral in parentheses immediately following the metal name:

- In $[\mathrm{Cu(CN)_4}]^{3-}$: $x + 4(-1) = -3 \implies x = +1 \implies \mathrm{Cu(I)}$.
- In $[\mathrm{Fe(CN)_6}]^{4-}$: $x + 6(-1) = -4 \implies x = +2 \implies \mathrm{Fe(II)}$.
- In $[\mathrm{Co(NH_3)_4(H_2O)Cl}]^{2+}$: $x + 4(0) + 1(0) + 1(-1) = +2 \implies x = +3 \implies \mathrm{Co(III)}$.
- In $[\mathrm{Ni(CO)_4}]$: $x + 4(0) = 0 \implies x = 0 \implies \mathrm{Ni(0)}$.

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(f) Homoleptic vs. Heteroleptic Complexes

- **Homoleptic Complexes:** Complexes in which the central metal atom/ion is coordinated to only *one kind* of donor group. Examples: $[Co(NH_3)_6]^{3+}$, $[Ni(CO)_4]$, $[Fe(CN)_6]^{3-}$.
- **Heteroleptic Complexes:** Complexes in which the central metal atom/ion is bound to *more than one kind* of donor group. Examples: $[Co(NH_3)_4Cl_2]^{+}$, $[PtCl_2(NH_3)_2]$. (Heteroleptic complexes can exhibit geometrical and optical isomerism).

4. IUPAC Nomenclature of Mononuclear Coordination Compounds

To communicate structures and isomerism unambiguously, the International Union of Pure and Applied Chemistry (IUPAC) has established standardized principles of additive nomenclature for coordination entities.

IUPAC Rules for Writing Chemical Formulas

1. The central metal atom is placed first inside the square brackets.
2. Ligands are then listed in **strict alphabetical order** according to their chemical symbols/abbreviations, regardless of their ionic charge. (In case of abbreviations like 'en' or 'ox', the first letter of the abbreviation determines its alphabetical position).
3. The entire coordination entity is enclosed in square brackets $[\ldots]$. Polyatomic ligands and abbreviations are individually enclosed in parentheses, e.g., $[Co(NH_3)_4(H_2O)Cl]Cl_2$.
4. No space is left between the central metal symbol and the ligands inside the square brackets.
5. When writing a charged entity without its counter ion, the net charge is indicated outside the square brackets as a right superscript with the numeral preceding the sign: $[Cr(H_2O)_6]^{3+}$, $[Co(CN)_6]^{3-}$.
6. The total charge of the cations must equal and balance the total charge of the anions.

IUPAC Rules for Systematic Naming

1. **Order of Naming Ions:** The positive ion (cation) is always named first, followed by the negative ion (anion), regardless of whether the complex entity is cationic, anionic, or neutral.
2. **Ligand Order:** Ligands are named in strict alphabetical order before the name of the central metal atom/ion. (Numerical multipliers such as di-, tri-, tetra- are ignored when determining alphabetical order).
3. **Ligand Name Endings:**
   • *Anionic ligands:* End in **-o** (or **-ido** according to 2004 IUPAC recommendations): chloride → *chlorido* (or chloro), cyanide → *cyanido*, hydroxide → *hydroxido*, oxalate → *oxalato*, sulphate → *sulphato*, nitrite → *nitrito-N* (or nitro) or *nitrito-O*.
   • *Neutral ligands:* Retain their molecular names, with four key exceptions: H_2O is named **aqua**, NH_3 is named **ammine** (note the double 'm'), CO is named **carbonyl**, and NO is named **nitrosyl**.
4. **Numerical Prefixes:**
   • Simple ligands take prefixes *mono-*, *di-*, *tri-*, *tetra-*, *penta-*, *hexa-*.
   • Complex ligands (especially those whose names already contain a numerical prefix, e.g., ethane-1,2-diamine, triphenylphosphine) or polydentate chelating ligands take multiplicative prefixes **bis-** (2), **tris-** (3), **tetrakis-** (4), with the ligand name placed in parentheses: e.g., $\mathrm{tris(ethane-1,2-diamine)cobalt(III)}$.
5. **Central Metal Name:**
   • *In a Cationic or Neutral Entity:* The metal retains its regular English elemental name: cobalt, chromium, platinum, iron, nickel, copper.

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• *In an Anionic Entity:* The metal name ends in the characteristic suffix **-ate**. For certain metals, the Latin root is used: iron → **ferrate**, copper → **cuprate**, lead → **plumbate**, silver → **argentate**, gold → **aurate**, tin → **stannate**, platinum → **platinate**, cobalt → **cobaltate**.

6. **Oxidation State:** The oxidation state of the central metal is indicated by a Roman numeral in parentheses immediately following the metal name, with no space: e.g., $\mathbf{cobalt(III)}$, $\mathbf{nickel(0)}$.

Comprehensive Master Table of IUPAC Nomenclature

Formula	Type of Entity	Central Metal & Oxidation State	Systematic IUPAC Name
$[Co(NH_3)_6]Cl_3$	Cationic complex	$Co(III)$	Hexaamminecobalt(III) chloride
$[CoCl(NH_3)_5]Cl_2$	Cationic complex	$Co(III)$	Pentaamminechloridocobalt(III) chloride
$K_4[Fe(CN)_6]$	Anionic complex	$Fe(II)$	Potassium hexacyanidoferrate(II)
$K_3[Fe(CN)_6]$	Anionic complex	$Fe(III)$	Potassium hexacyanidoferrate(III)
$K_3[Cr(C_2O_4)_3]$	Anionic complex	$Cr(III)$	Potassium trioxalatochromate(III)
$[PtCl_2(en)_2]Cl_2$	Cationic complex	$Pt(IV)$	Dichloridobis(ethane-1,2-diamine)platinum(IV) chloride
$[Ni(CO)_4]$	Neutral complex	$Ni(0)$	Tetracarbonylnickel(0)
$[Co(NH_3)_3(NO_2)_3]$	Neutral complex	$Co(III)$	Triamminetrinitrito-N-cobalt(III)
$[Cr(NH_3)_3(H_2O)_3]Cl_3$	Cationic complex	$Cr(III)$	Triamminetriaquachromium(III) chloride
$[Ag(NH_3)_2][Ag(CN)_2]$	Cationic + Anionic complex	$Ag(I)$ & $Ag(I)$	Diamminesilver(I) dicyanidoargentate(I)
$Hg[Co(SCN)_4]$	Anionic complex	$Co(II)$ (with $Hg(II)$)	Mercury(II) tetrathiocyanato-S-cobaltate(II)
$Fe_4[Fe(CN)_6]_3$	Anionic complex (Prussian blue)	Counter: $Fe(III)$; Complex: $Fe(II)$	Iron(III) hexacyanidoferrate(II)

5. Isomerism in Coordination Compounds

Isomers are two or more compounds that possess identical chemical formulas but display different structural arrangements of atoms or differing spatial orientations, resulting in distinct physical and chemical properties.

5.1 Structural Isomerism

Structural isomers possess identical overall chemical formulas but differ in the connectivity of chemical bonds within the coordination sphere and counter ions:

- **1. Ionisation Isomerism:** Occurs when an ionisable counter ion displaces a ligand from within the coordination sphere, exchanging roles. When dissolved in water, the two isomers produce distinctly different ions in solution.
  • *Diagnostic chemical test:*

$$[Co(NH_3)_5(SO_4)]Br \xrightarrow{BaCl_2} \text{No precipitate} \quad \Big| \quad [Co(NH_3)_5(SO_4)]Br \xrightarrow{AgNO_3} AgBr\downarrow \ (\text{pale yellow precipitate})$$

$$[Co(NH_3)_5Br]SO_4 \xrightarrow{BaCl_2} BaSO_4\downarrow \ (\text{white precipitate}) \quad \Big| \quad [Co(NH_3)_5Br]SO_4 \xrightarrow{AgNO_3} \text{No precipitate}$$

- **2. Solvate (Hydrate) Isomerism:** A specific form of ionisation isomerism where water acts as either a coordinated ligand within the coordination sphere or as uncoordinated solvent of crystallization held in the crystal lattice.
  • $[Cr(H_2O)_6]Cl_3$ (Violet): 6 coordinated waters; gives $3\ \text{mol}\ AgCl$ per mol of complex.

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• $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (Grey-green): 5 coordinated waters; gives $2\ mol\ AgCl$ per mol of complex.
• $[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$ (Bright green): 4 coordinated waters; gives $1\ mol\ AgCl$ per mol of complex.

- **3. Linkage Isomerism:** Arises exclusively in complexes containing *ambidentate ligands* that possess two different donor atoms:
  • Pentaamminenitritocobalt(III) chloride exists as two distinct linkage isomers discovered by S. M. Jørgensen:
  – $[Co(NH_3)_5(NO_2)]Cl_2$ (Yellow form): Nitro isomer coordinated through nitrogen ($Co-NO_2$).
  – $[Co(NH_3)_5(ONO)]Cl_2$ (Red form): Nitrito isomer coordinated through oxygen ($Co-ONO$).
  • Thiocyanate complexes: $M-SCN$ (thiocyanato-S) vs. $M-NCS$ (thiocyanato-N or isothiocyanato).
- **4. Coordination Isomerism:** Occurs in salts where both the cation and the anion are complex entities, arising from the interchange of ligands between the two metal centers.
  • Example: $[Co(NH_3)_6][Cr(CN)_6]$ (where NH_3 coordinates to $Co(III)$ and CN^- to $Cr(III)$) and its coordination isomer $[Cr(NH_3)_6][Co(CN)_6]$ (where NH_3 coordinates to $Cr(III)$ and CN^- to $Co(III)$).

5.2 Stereoisomerism

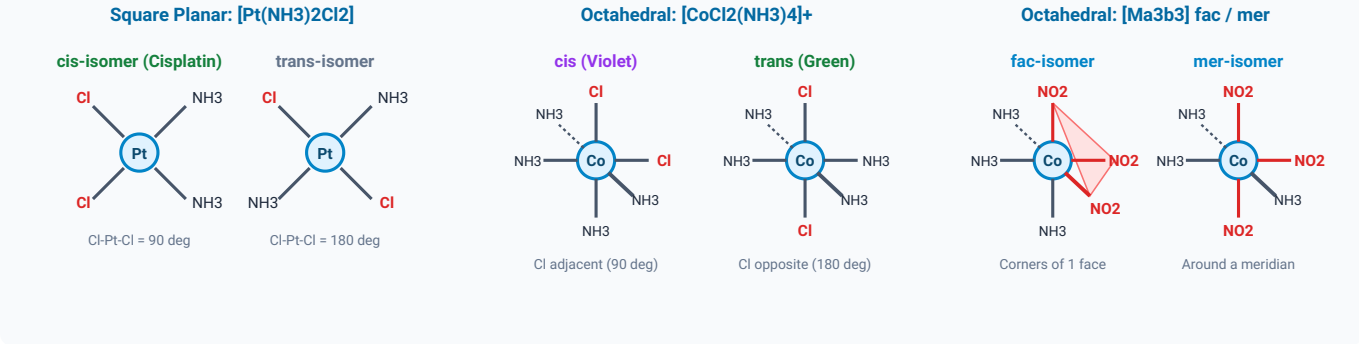
Stereoisomers possess identical chemical bonds and connectivity, but differ in the spatial orientation of these bonds.

A. Geometrical Isomerism

Arises in heteroleptic complexes due to different possible spatial distributions of coordinating ligands:

- **Square Planar Complexes ($CN=4$):**
  • *Type* $[MA_2B_2]$: Exhibits 2 geometrical isomers: **cis** (identical donor atoms occupy adjacent positions at $90°$) and **trans** (identical donor atoms occupy opposite positions at $180°$).
  – Classic example: $[PtCl_2(NH_3)_2]$. The *cis*-isomer (Cisplatin) is an effective antitumor drug, whereas the *trans*-isomer is clinically inactive.
  • *Type* $[MABCD]$: A square planar complex containing four different monodentate ligands (e.g., $[Pt(NH_3)(Br)(Cl)(py)]$) exhibits **3 geometrical isomers** (two cis and one trans with respect to any chosen reference ligand).
- **Why Tetrahedral Complexes Cannot Show Geometrical Isomerism:** In a regular tetrahedral geometry, all four coordination positions are mutually adjacent and equidistant from one another, subtending identical bond angles of $109.5°$. Therefore, no *cis*/*trans* distinction is geometrically possible.
- **Octahedral Complexes ($CN=6$):**
  • *Type* $[MA_4B_2]$: Shows 2 isomers: **cis** (the two B ligands are adjacent at $90°$) and **trans** (the two B ligands are opposite at $180°$). Example: $[CoCl_2(NH_3)_4]^+$ (cis is violet, trans is green).
  • *Type* $[MA_3B_3]$: Displays a distinctive pair of geometrical isomers:
  – **Facial (fac) Isomer:** The three identical donor atoms occupy adjacent positions at the corners of a single triangular octahedral face (all mutually adjacent at $90°$).
  – **Meridional (mer) Isomer:** The three identical donor atoms lie in a plane that passes through the metal center, tracing a meridian around the octahedron (two ligands trans at $180°$, one cis at $90°$). Example: $[Co(NH_3)_3(NO_2)_3]$.
  • *Type* $[M(AA)_2B_2]$: Complexes with two symmetrical didentate chelating ligands and two monodentate ligands exhibit *cis* and *trans* forms, such as in $[CoCl_2(en)_2]^+$.

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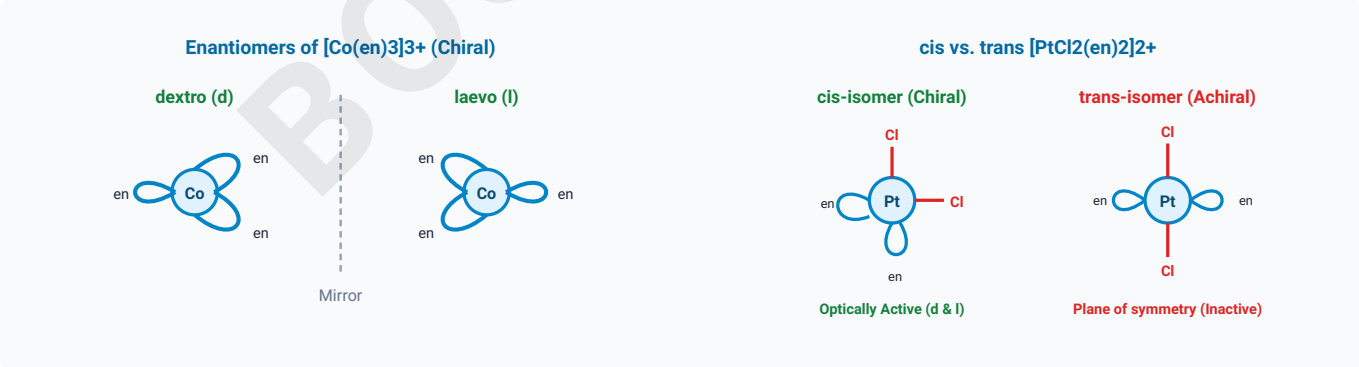


Tutorial Figure 3: Geometrical Isomerism in Square Planar ([MA2B2]) and Octahedral Complexes ([MA4B2] cis/trans and [MA3B3] fac/mer).

B. Optical Isomerism

Optical isomers (enantiomers) are stereoisomers that are non-superimposable mirror images of one another. Molecules or ions that lack an improper axis of rotation (specifically lacking a plane of symmetry σ and a center of inversion i) are **chiral**.

- Chiral coordination entities rotate the plane of plane-polarized light in a polarimeter: the isomer rotating the plane to the right is **dextrorotatory (*d*)**, while that rotating to the left is **laevorotatory (*l*)**.
- **Tris-chelate complexes of the type $[\mathrm{M(AA)_3}]^{n+}$:** (e.g., $[\mathrm{Co(en)_3}]^{3+}$, $[\mathrm{Cr(C_2O_4)_3}]^{3-}$) have no plane of symmetry; they are universally chiral and exist as an enantiomeric pair of *d*- and *l*-forms. They show *no* geometrical isomerism.
- **Bis-chelate complexes of the type $[\mathrm{M(AA)_2B_2}]^{n+}$:** (e.g., $[\mathrm{CoCl_2(en)_2}]^{+}$, $[\mathrm{PtCl_2(en)_2}]^{2+}$, $[\mathrm{CrCl_2(ox)_2}]^{3-}$):
  • *cis-isomer:* Has no plane of symmetry or inversion center. Its mirror image cannot be superimposed; thus, **cis is chiral and optically active** (resolvable into *d* and *l* enantiomers).
  • *trans-isomer:* Possesses both a plane of symmetry and a center of inversion. It is superimposable on its mirror image; thus, **trans is achiral and optically inactive**.



Tutorial Figure 4: Optical Isomerism in Octahedral Complexes. Tris-chelate [Co(en)3]3+ exists as non-superimposable d and l enantiomers; in [PtCl2(en)2]2+, only the cis-isomer is chiral, while the trans-isomer possesses a plane of symmetry and is optically inactive.

6. Bonding in Coordination Compounds: Valence Bond Theory (VBT)

Linus Pauling extended the Valence Bond Theory to explain the geometries and magnetic properties of coordination entities. According to VBT, under the influence of approaching ligands, the central metal atom or ion makes available empty valence atomic orbitals that undergo **hybridisation** to yield an equivalent set of

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hybrid orbitals directed along the symmetry axes of the coordination polyhedron. Each ligand then donates a lone pair of electrons into an empty hybrid orbital to establish a metal–ligand coordinate covalent σ-bond.

Coordination Number	Hybridisation Scheme	Spatial Geometry	Representative Complexes
4	sp^3	Tetrahedral	$[NiCl_4]^{2-}$, $[Ni(CO)_4]$, $[MnBr_4]^{2-}$, $[Zn(NH_3)_4]^{2+}$
4	dsp^2	Square planar	$[Ni(CN)_4]^{2-}$, $[PtCl_4]^{2-}$, $[Pt(NH_3)_4]^{2+}$, $[Cu(NH_3)_4]^{2+}$
5	sp^3d	Trigonal bipyramidal	$[Fe(CO)_5]$
6	d^2sp^3	Octahedral (Inner orbital / Low spin)	$[Co(NH_3)_6]^{3+}$, $[Fe(CN)_6]^{4-}$, $[Fe(CN)_6]^{3-}$, $[Cr(NH_3)_6]^{3+}$
6	sp^3d^2	Octahedral (Outer orbital / High spin)	$[CoF_6]^{3-}$, $[FeF_6]^{3-}$, $[MnCl_6]^{3-}$, $[Ni(NH_3)_6]^{2+}$

Magnetic Properties and the Spin-Only Formula

The magnetic susceptibility of a complex reveals the number of unpaired electrons (n) present in the metal d-orbitals. The theoretical **spin-only magnetic moment** (μ_s) is calculated as:

$$\mu_s = \sqrt{n(n+2)} \text{ Bohr Magnetons (BM)}$$

Key reference numerical values for NEET:

- $n = 1 \text{ unpaired } e^- \implies \mu_s = \sqrt{1(3)} = 1.73 \text{ BM}$
- $n = 2 \text{ unpaired } e^- \implies \mu_s = \sqrt{2(4)} = 2.83 \text{ BM}$
- $n = 3 \text{ unpaired } e^- \implies \mu_s = \sqrt{3(5)} = 3.87 \text{ BM}$
- $n = 4 \text{ unpaired } e^- \implies \mu_s = \sqrt{4(6)} = 4.90 \text{ BM}$
- $n = 5 \text{ unpaired } e^- \implies \mu_s = \sqrt{5(7)} = 5.92 \text{ BM}$

BOSONANCE

Case 1: Inner Orbital vs. Outer Orbital Cobalt(III) Complexes ($3d^6$)

Cobalt in ground state: $[\mathrm{Ar}]\,3d^7 4s^2$. For Co^{3+}: $[\mathrm{Ar}]\,3d^6 4s^0 4p^0 4d^0$.

- $[\mathrm{Co(NH_3)_6}]^{3+}$ **(Diamagnetic, Inner Orbital):**
  In the presence of the strong-field ligand $\mathrm{NH_3}$, the six electrons in the $3d$ subshell are forced to pair up completely:

  $$3d^6 \rightarrow [\uparrow\downarrow][\uparrow\downarrow][\uparrow\downarrow][\quad][\quad]$$

  This vacates two inner $3d$ orbitals. The two vacant $3d$, one $4s$, and three $4p$ orbitals hybridize to form six equivalent d^2sp^3 hybrid orbitals. Six lone pairs from six $\mathrm{NH_3}$ ligands fill these hybrid orbitals.
  Magnetic State: Diamagnetic ($n = 0$, $\mu = 0\ \mathrm{BM}$). Because inner $(n-1)d$ orbitals are utilized, it is termed an **inner orbital complex**, **low spin complex**, or **spin-paired complex**.
- $[\mathrm{CoF_6}]^{3-}$ **(Paramagnetic, Outer Orbital):**
  In the presence of the weak-field ligand F^-, electron pairing in the $3d$ subshell cannot be induced:

  $$3d^6 \rightarrow [\uparrow\downarrow][\uparrow][\uparrow][\uparrow][\uparrow] \quad (n = 4)$$

  No inner $3d$ orbitals are available. The metal ion instead hybridizes its outer $4s$, three $4p$, and two outer $4d$ orbitals to form six sp^3d^2 hybrid orbitals.
  Magnetic State: Strongly paramagnetic ($n = 4$, $\mu = \sqrt{4(6)} = 4.90\ \mathrm{BM}$). Because outer nd orbitals are utilized, it is termed an **outer orbital complex**, **high spin complex**, or **spin-free complex**.

BOSONANCE

Case 2: Four-Coordinate Nickel Complexes ($3d^8$)

Nickel in ground state: $[\mathrm{Ar}]\,3d^8 4s^2$. For Ni^{2+}: $[\mathrm{Ar}]\,3d^8 4s^0 4p^0$.

- $[\mathrm{NiCl_4}]^{2-}$ **(Tetrahedral, Paramagnetic):**
  Cl^- is a weak-field ligand unable to pair the two unpaired electrons in $3d^8$:

  $$3d^8 \rightarrow [\uparrow\downarrow][\uparrow\downarrow][\uparrow\downarrow][\uparrow][\uparrow] \quad (n = 2)$$

  The vacant $4s$ and three $4p$ orbitals undergo sp^3 hybridisation, yielding a **tetrahedral geometry**.
  Magnetic Moment: $\mu = \sqrt{2(4)} = 2.83\ \mathrm{BM}$ (paramagnetic).
- $[\mathrm{Ni(CN)_4}]^{2-}$ **(Square Planar, Diamagnetic):**
  CN^- is a strong-field ligand that forces pairing of the two unpaired $3d$ electrons:

  $$3d^8 \rightarrow [\uparrow\downarrow][\uparrow\downarrow][\uparrow\downarrow][\uparrow\downarrow][\quad] \quad (n = 0)$$

  One inner $3d$ orbital is vacated. The vacant $3d$, one $4s$, and two $4p$ orbitals undergo dsp^2 hybridisation, creating a **square planar geometry**.
  Magnetic Moment: Diamagnetic ($n = 0$, $\mu = 0\ \mathrm{BM}$).
- $[\mathrm{Ni(CO)_4}]$ **(Tetrahedral, Diamagnetic, $\mathrm{Ni}(0)$):**
  In zero oxidation state, nickel has the configuration $3d^8 4s^2$. Under the influence of the extraordinarily strong ligand CO, the two $4s$ electrons are pushed completely into the $3d$ subshell, pairing with the unpaired electrons to generate a fully filled $3d^{10}$ configuration:

  $$3d^{10} \rightarrow [\uparrow\downarrow][\uparrow\downarrow][\uparrow\downarrow][\uparrow\downarrow][\uparrow\downarrow] \quad 4s^0 \quad 4p^0$$

  The now vacant $4s$ and three $4p$ orbitals undergo sp^3 hybridisation, producing a **tetrahedral geometry**.
  Magnetic Moment: Diamagnetic ($n = 0$, $\mu = 0\ \mathrm{BM}$).

Limitations of Valence Bond Theory

Although VBT provides a qualitative model for geometries and spin states, it has severe limitations:

- It involves multiple empirical assumptions and offers **no quantitative interpretation** of magnetic susceptibility data.
- It cannot explain or predict the **electronic absorption spectra and characteristic colors** of transition metal complexes.
- It gives no quantitative interpretation of the **thermodynamic or kinetic stabilities** of complexes.
- It cannot reliably predict whether 4-coordinate d^8 complexes (such as Cu^{2+} or Ni^{2+}) will adopt tetrahedral or square planar geometry.
- It provides no physical mechanism to explain **why some ligands are strong-field (pairing) while others are weak-field**.

7. Crystal Field Theory (CFT)

Crystal Field Theory (CFT) is an electrostatic model that treats the metal–ligand interaction as purely ionic, arising entirely from electrostatic attractions between the positively charged central metal cation and the negative point charges (anionic ligands like Cl^-, CN^-) or point dipoles (neutral dipolar molecules like $\mathrm{NH_3}$, $\mathrm{H_2O}$ directed with their negative ends toward the metal).

BOSONANCE

In an isolated, gaseous transition metal atom or ion, the five d-orbitals (d_{xy}, d_{yz}, d_{xz}, $d_{x^2-y^2}$, d_{z^2}) are completely **degenerate** (possessing identical energy). When surrounded by a hypothetical spherically symmetrical field of negative charges, all five d-orbitals are raised equally in energy. However, in an actual complex, the ligands approach along specific geometric directions. The electrostatic repulsion between ligand electrons and metal d-electrons is non-uniform, lifting the degeneracy and causing the d-orbitals to split into sets of differing energies. This phenomenon is termed **crystal field splitting**.

7.1 Crystal Field Splitting in Octahedral Complexes (Δ_o)

In an octahedral coordination entity, six ligands approach the central metal ion along the Cartesian axes (x, y, and z):

- **Axial Orbitals (e_g set):** The $d_{x^2-y^2}$ and d_{z^2} orbitals have their electron density lobes pointing directly along the Cartesian axes. Consequently, they experience maximum electrostatic repulsion from the incoming ligands and are destabilized, rising in energy by $+\frac{3}{5}\Delta_o$ $(+0.6\Delta_o)$ above the spherical barycentre. The doubly degenerate set is designated e_g.
- **Non-Axial Orbitals (t_{2g} set):** The d_{xy}, d_{yz}, and d_{xz} orbitals have their lobes directed between the Cartesian axes (at 45° to the axes). They experience significantly less repulsion and are stabilized, dropping in energy by $-\frac{2}{5}\Delta_o$ $(-0.4\Delta_o)$ below the barycentre. The triply degenerate set is designated t_{2g}.

The total energy separation between the t_{2g} and e_g levels is defined as the **octahedral crystal field splitting energy**, denoted by Δ_o (or $10\,Dq$).

7.2 Crystal Field Splitting in Tetrahedral Complexes (Δ_t)

In a tetrahedral complex, four ligands approach along the alternate corners of a circumscribed cube. The coordinate axes pass through the centers of the cube faces. Hence, the four ligands approach *between* the Cartesian axes:

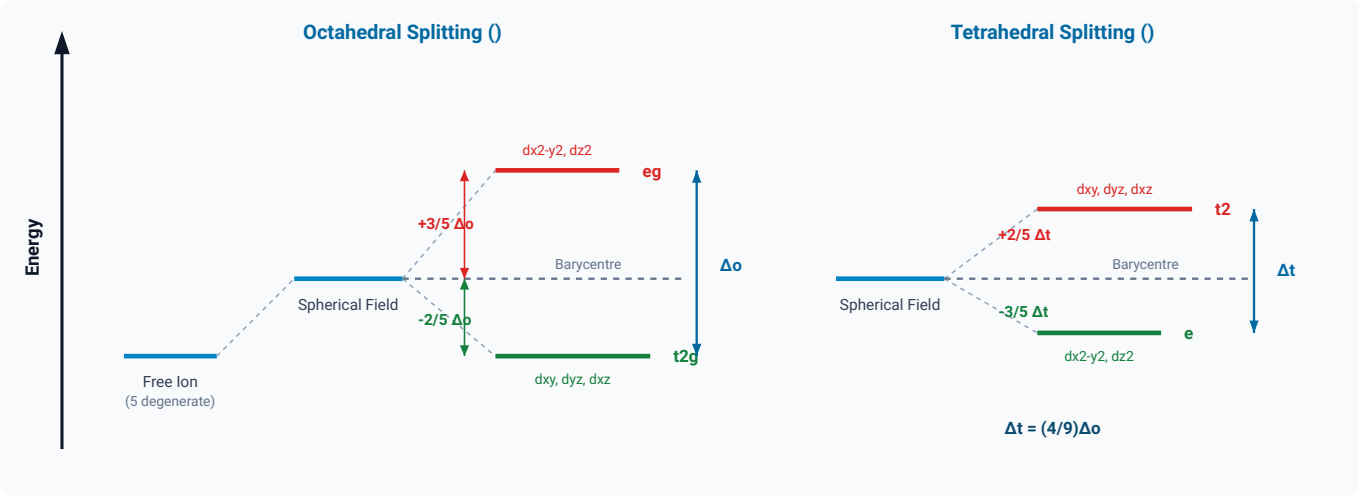
- The non-axial orbitals (d_{xy}, d_{yz}, d_{xz}) point more closely toward the approaching ligands than the axial orbitals, experiencing greater repulsion. Therefore, the t_2 set is destabilized by $+\frac{2}{5}\Delta_t$ $(+0.4\Delta_t)$.
- The axial orbitals ($d_{x^2-y^2}$, d_{z^2}) point between the ligands and are stabilized by $-\frac{3}{5}\Delta_t$ $(-0.6\Delta_t)$.
- *Note on symmetry labels:* Because a regular tetrahedron lacks a center of inversion symmetry (centrosymmetry), the 'g' (gerade) subscript is omitted: the levels are designated strictly as e (lower energy, doubly degenerate) and t_2 (higher energy, triply degenerate).

NEET Fundamental Formula: For the same metal ion, same ligands, and identical metal–ligand bond lengths:

$$\Delta_t = \frac{4}{9}\Delta_o$$

Because Δ_t is less than half of Δ_o, the crystal field splitting in tetrahedral complexes is *never large enough to overcome the electron pairing energy* ($\Delta_t < P$ universally). Consequently, **tetrahedral complexes are virtually always high-spin (spin-free)**!

BOSONANCE



Tutorial Figure 5: Complete Crystal Field Splitting Diagrams for Octahedral and Tetrahedral Geometries. Notice the inverted splitting pattern and the absence of 'g' subscripts in tetrahedral complexes.

7.3 The Spectrochemical Series

The magnitude of Δ_o depends on both the oxidation state of the metal ion and the field strength of the coordinated ligands. Ligands that produce large Δ_o splitting are termed **strong field ligands**, while those producing small splitting are **weak field ligands**. Arranged in order of increasing crystal field splitting (experimentally derived from absorption spectra), they constitute the **spectrochemical series**:

$$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$$

NEET Memory Anchor: Halides are weak field ligands → Oxygen-donor ligands are intermediate-to-weak → Nitrogen-donor ligands are strong → Carbon-donor ligands (CN^-, CO) are the strongest field ligands.

7.4 Electronic Configurations & Crystal Field Stabilization Energy (CFSE)

In an octahedral field, the first three electrons enter the lower-energy t_{2g} orbitals singly in accordance with Hund's rule: $d^1 \rightarrow t_{2g}^1$, $d^2 \rightarrow t_{2g}^2$, $d^3 \rightarrow t_{2g}^3$. For a d^4 ion, a competition arises between the crystal field splitting energy (Δ_o) and the **mean pairing energy (P)** (the electrostatic repulsion energy required to pair two electrons in the same orbital):

- **Case 1: Weak Field Ligands ($\Delta_o < P$):**
  The cost of pairing (P) exceeds the energy gap (Δ_o). The fourth electron enters the higher-energy e_g orbital:

$$\text{Configuration: } t_{2g}^3 e_g^1 \quad \text{(High Spin / Spin Free)}$$

- **Case 2: Strong Field Ligands ($\Delta_o > P$):**
  The energy gap (Δ_o) is larger than the pairing energy (P). The fourth electron pairs with an electron in a t_{2g} orbital:

$$\text{Configuration: } t_{2g}^4 e_g^0 \quad \text{(Low Spin / Spin Paired)}$$

System	High Spin ($\Delta_o < P$)	Unpaired e^- (n)	Low Spin ($\Delta_o > P$)	Unpaired e^- (n)
d^1	$t_{2g}^1 e_g^0$	1	$t_{2g}^1 e_g^0$	1
d^2	$t_{2g}^2 e_g^0$	2	$t_{2g}^2 e_g^0$	2

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System	High Spin ( )	Unpaired ( )	Low Spin ( )	Unpaired ( )
d^3	$t_{2g}^3 e_g^0$	3	$t_{2g}^3 e_g^0$	3
d^4	$t_{2g}^3 e_g^1$	4	$t_{2g}^4 e_g^0$	2
d^5	$t_{2g}^3 e_g^2$	5	$t_{2g}^5 e_g^0$	1
d^6	$t_{2g}^4 e_g^2$	4	$t_{2g}^6 e_g^0$	0 (diamagnetic)
d^7	$t_{2g}^5 e_g^2$	3	$t_{2g}^6 e_g^1$	1
d^8	$t_{2g}^6 e_g^2$	2	$t_{2g}^6 e_g^2$	2
d^9	$t_{2g}^6 e_g^3$	1	$t_{2g}^6 e_g^3$	1
d^{10}	$t_{2g}^6 e_g^4$	0	$t_{2g}^6 e_g^4$	0

Formula for CFSE in Octahedral Complexes:

$$\mathrm{CFSE_{oct}} = [-0.4\,n(t_{2g}) + 0.6\,n(e_g)]\Delta_o + m\mathrm{P}$$

where $n(t_{2g})$ and $n(e_g)$ are the numbers of electrons in the respective levels, and m is the number of extra electron pairs formed over the free-ion ground state.

7.5 Origin of Colour and $d-d$ Transitions

Transition metal complexes frequently exhibit vivid colours. When white light passes through a solution of a complex, a photon whose energy corresponds precisely to the crystal field splitting ($h\nu = \Delta_o$) is absorbed, promoting an electron from the lower t_{2g} level to the empty or half-filled e_g level (a $d-d$ **electronic transition**). The transmitted light consists of the residual wavelengths, producing the **complementary colour**:

Coordination Entity	Wavelength Absorbed (nm)	Colour Absorbed	Observed Colour of Complex
$[CoCl(NH_3)_5]^{2+}$	535 nm	Yellow	Violet
$[Co(NH_3)_5(H_2O)]^{3+}$	500 nm	Blue-Green	Red
$[Co(NH_3)_6]^{3+}$	475 nm	Blue	Yellow-Orange
$[Co(CN)_6]^{3-}$	310 nm	Ultraviolet (invisible)	Pale Yellow
$[Cu(H_2O)_4]^{2+}$	600 nm	Red	Blue
$[Ti(H_2O)_6]^{3+}$	498 nm	Blue-Green	Violet

NEET High-Yield Principles of Complex Colour

- **Requirement for $d-d$ Transitions:** In the total absence of ligands, no crystal field splitting occurs; the d-orbitals remain degenerate, preventing $d-d$ absorption. Thus, anhydrous $CuSO_4$ is white, whereas hydrated $CuSO_4 \cdot 5H_2O$ is blue. Heating $[Ti(H_2O)_6]Cl_3$ drives off water ligands, rendering it completely colourless.
- **d^0 and d^{10} Configurations:** Ions with empty d-orbitals (d^0, e.g., Sc^{3+}, Ti^{4+}) or fully occupied d-orbitals (d^{10}, e.g., Zn^{2+}, Cd^{2+}) cannot undergo $d-d$ transitions and are universally colourless.
- **Colour Tuning by Ligand Substitution:** Adding ethane-1,2-diamine progressively to green $[Ni(H_2O)_6]^{2+}$ causes a stepwise increase in Δ_o:

$$[Ni(H_2O)_6]^{2+}\ (\text{green}) \xrightarrow{+en} [Ni(H_2O)_4(en)]^{2+}\ (\text{pale blue}) \xrightarrow{+en} [Ni(H_2O)_2(en)_2]^{2+}\ (\text{blue/purple}) \xrightarrow{+en} [Ni(en)_3]^{2+}\ (\text{violet})$$

- **Gemstones:**
  • *Ruby:* Formed by mineral corundum (Al_2O_3) containing $0.5\text{--}1\%$ Cr^{3+} ($3d^3$) substituting in octahedral Al^{3+} sites. Strong $d-d$ transitions absorb green/yellow light, imparting a brilliant red

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color.
• *Emerald:* Mineral beryl ($\mathrm{Be_3Al_2Si_6O_{18}}$) where $\mathrm{Cr^{3+}}$ occupies octahedral sites with slightly larger metal–ligand distances (smaller Δ_o), shifting the absorption bands to yellow-red and blue, thus transmitting vibrant green light.

7.6 Limitations of Crystal Field Theory

- Because CFT treats ligands as negative point charges, it incorrectly predicts that **anionic ligands should exert the largest splitting**. In reality, halides are at the lowest (weak-field) end of the spectrochemical series.
- It completely ignores the **covalent character** of metal–ligand bonds, neglecting π-bonding and orbital overlap (addressed in Ligand Field and Molecular Orbital theories).

8. Bonding and Structure in Metal Carbonyls

Transition metals form a distinct class of homoleptic compounds with carbon monoxide (CO) called **metal carbonyls**, wherein the metal center is formally in the zero oxidation state. These compounds possess well-defined molecular geometries:

- $[\mathrm{Ni(CO)_4}]$: Mononuclear, tetrahedral geometry ($\mathrm{Ni(0)}$, sp^3).
- $[\mathrm{Fe(CO)_5}]$: Mononuclear, trigonal bipyramidal geometry ($\mathrm{Fe(0)}$, dsp^3).
- $[\mathrm{Cr(CO)_6}]$: Mononuclear, octahedral geometry ($\mathrm{Cr(0)}$, d^2sp^3).
- $[\mathrm{Mn_2(CO)_{10}}]$: Dinuclear decacarbonyl, composed of two square pyramidal $\mathrm{Mn(CO)_5}$ units joined by a direct $\mathrm{Mn-Mn}$ single bond (bond length $292\ \mathrm{pm}$), without bridging carbonyls.
- $[\mathrm{Co_2(CO)_8}]$: Dinuclear octacarbonyl; in the solid state it features a $\mathrm{Co-Co}$ single bond bridged by two carbonyl ligands, with each cobalt bearing three terminal CO groups.

The Synergic Bonding Mechanism

The metal–carbon ($\mathrm{M-C}$) bond in metal carbonyls exhibits unique mutual reinforcement through a two-way **synergic interaction** involving both σ and π bonding components:

Components of the Synergic Bond

1. $\mathrm{M \leftarrow C}$ **σ-Bond:** Formed by the donation of the lone pair of electrons residing in the non-bonding sp-hybridized orbital on the carbonyl carbon into a vacant d-orbital of the metal atom.
2. $\mathrm{M \rightarrow C}$ **π-Backbond:** Formed by the back-donation of a pair of electrons from a filled metal d-orbital of suitable π-symmetry into the empty, lowest unoccupied molecular orbital (LUMO)—the *antibonding π^* molecular orbital* of carbon monoxide.

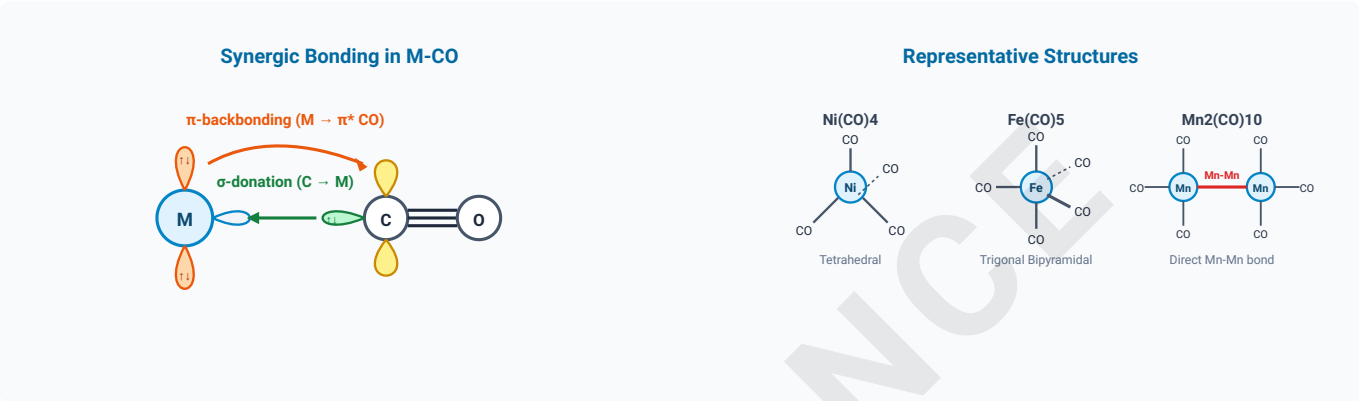
NEET High-Yield Deductions on Synergic Bonding

- **Mutual Strengthening:** The forward σ-donation builds up negative electron density on the metal, which enhances the metal's tendency to back-donate into the π^* orbitals of CO. This synergic effect dramatically strengthens and shortens the $\mathrm{M-C}$ bond.
- **Consequence on the Carbon–Oxygen ($\mathrm{C-O}$) Bond:** Because electrons are accepted into the *antibonding π^* orbital* of CO, the $\mathrm{C-O}$ bond order decreases below 3. Consequently:
  • The $\mathrm{C-O}$ bond length *increases* compared to free CO.
  • The $\mathrm{C-O}$ stretching frequency (ν_{CO}) *decreases*.

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- **Influence of Metal Charge (Isoelectronic Carbonyls):**
  Consider the isoelectronic series $[V(CO)_6]^-$, $[Cr(CO)_6]$, and $[Mn(CO)_6]^+$:
  • The negatively charged complex $[V(CO)_6]^-$ possesses the highest electron density on the metal center $\rightarrow$ exhibits the *greatest* π*-backbonding* $\rightarrow$ has the **strongest** $M-C$ **bond** and the **weakest / longest** $C-O$ **bond** (lowest ν_{CO}).

  • The positively charged complex $[Mn(CO)_6]^+$ holds its electrons tightly $\rightarrow$ displays the *least* π*-backbonding* $\rightarrow$ has the **weakest** $M-C$ **bond** and the **strongest / shortest** $C-O$ **bond** (highest ν_{CO}).



Tutorial Figure 6: The Synergic Bonding Mechanism in Transition Metal Carbonyls and Representative Homoleptic Structures.

9. Importance and Practical Applications of Coordination Compounds

Coordination compounds play foundational roles across chemical analysis, extractive metallurgy, biological life processes, industrial catalysis, and pharmacology.

1. Analytical Chemistry & Water Hardness

- **Complexometric Titrations:** Estimation of total water hardness caused by dissolved Ca^{2+} and Mg^{2+} ions is carried out by volumetric titration with standard disodium EDTA (Na_2EDTA). Selective estimation is achievable because the stability constant of $[Ca(EDTA)]^{2-}$ is significantly greater than that of $[Mg(EDTA)]^{2-}$.
- **Qualitative Analysis Reagents:**
  • *Dimethylglyoxime (DMG):* Used for the specific detection and gravimetric estimation of nickel(II) ion in ammoniacal medium, forming a characteristic **scarlet red precipitate** of bis(dimethylglyoximato)nickel(II), stabilized by intramolecular hydrogen bonds.
  • Other analytical chelating reagents include α-nitroso-β-naphthol (for cobalt) and cupron (for copper).

2. Extractive Metallurgy and Metal Refining

- **Cyanide Extraction of Gold and Silver (Mac-Arthur Forrest Process):** Native gold and silver are selectively leached from their ores by aerated dilute aqueous potassium/sodium cyanide:

$$4\,Au + 8\,CN^- + 2\,H_2O + O_2 \rightarrow 4\,[Au(CN)_2]^- + 4\,OH^-$$

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The pure metal is subsequently displaced from the soluble dicyanidoaurate(I) complex by reduction with zinc dust:

$$2\,[\mathrm{Au(CN)_2}]^- + \mathrm{Zn} \rightarrow [\mathrm{Zn(CN)_4}]^{2-} + 2\,\mathrm{Au}\downarrow$$

- **Mond Process for Nickel Refining:** Impure nickel reacts reversibly with carbon monoxide at $330\text{--}350\,\mathrm{K}$ to yield volatile tetracarbonylnickel(0), which on heating to $450\text{--}470\,\mathrm{K}$ decomposes into pure nickel:

$$\mathrm{Ni} + 4\,\mathrm{CO} \xrightarrow{330-350\,\mathrm{K}} [\mathrm{Ni(CO)_4}] \xrightarrow{450-470\,\mathrm{K}} \mathrm{Ni\ (pure)} + 4\,\mathrm{CO}$$

3. Biological Systems

- **Chlorophyll:** The green photosynthetic pigment of plants is a coordination complex of **magnesium ($\mathrm{Mg^{2+}}$)** with a substituted porphyrin ring.
- **Haemoglobin:** The red oxygen-transport pigment in red blood cells is a coordination complex of **iron ($\mathrm{Fe^{2+}}$)** with a porphyrin ring (heme).
- **Vitamin $\mathrm{B_{12}}$ (Cyanocobalamin):** The anti-pernicious anaemia factor is a coordination complex of **cobalt ($\mathrm{Co^{3+}}$)** with a corrin ring.
- **Metalloenzymes:** Enzymes such as *carbonic anhydrase* (catalyzing $\mathrm{CO_2}$ hydration) and *carboxypeptidase A* (hydrolyzing peptide bonds) contain coordinated **zinc ($\mathrm{Zn^{2+}}$)** ions at their active sites.

4. Industrial Catalysis & Chemical Technology

- **Wilkinson's Catalyst:** Chloridotris(triphenylphosphine)rhodium(I), $[\mathrm{(Ph_3P)_3RhCl}]$, is a square planar complex utilized as a homogeneous catalyst for the selective **hydrogenation of alkenes** at ambient temperature and pressure.
- **Electroplating:** Articles are electroplated with silver or gold much more smoothly, adherent, and uniformly from baths containing $[\mathrm{Ag(CN)_2}]^-$ or $[\mathrm{Au(CN)_2}]^-$ than from solutions containing uncomplexed simple metal ions, because the low equilibrium concentration of free metal ions prevents rapid, uneven dendritic crystallization.
- **Black & White Photography:** The exposed and developed film is fixed by washing with hypo solution (aqueous sodium thiosulphate, $\mathrm{Na_2S_2O_3}$), which dissolves unexposed, insoluble silver bromide by forming the soluble complex ion:

$$\mathrm{AgBr} + 2\,\mathrm{S_2O_3^{2-}} \rightarrow [\mathrm{Ag(S_2O_3)_2}]^{3-} + \mathrm{Br^-}$$

5. Medicinal Chemistry & Chelation Therapy

- **Removal of Toxic Metals:**
  • *Lead Poisoning:* Chelation therapy with calcium disodium EDTA ($\mathrm{CaNa_2EDTA}$) removes toxic lead ions as stable, soluble, non-toxic lead-EDTA chelates excreted in urine.
  • *Copper Toxicity (Wilson's Disease):* Excess copper is chelated and eliminated using **D-penicillamine**.
  • *Iron Overload:* Excess iron is selectively scavenged by the hexadentate siderophore chelator **desferrioxime B**.
- **Antitumor Chemotherapy:** Platinum coordination complexes effectively inhibit the proliferation of malignant tumors. **Cisplatin**, *cis*-$[\mathrm{PtCl_2(NH_3)_2}]$, coordinates to nuclear DNA, cross-linking adjacent guanine bases to prevent DNA replication in rapidly dividing cancer cells. (The *trans*-isomer is pharmacologically inactive).

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10. Comprehensive Worked Examples (Step-by-Step Problem Solving)

Worked Example 1: Werner's Formulations & Precipitation Stoichiometry

Problem: On treating $0.01\ \text{mol}$ of each of the following coordination compounds with excess cold aqueous AgNO_3, the masses of AgCl precipitated were determined as follows:

- Compound A ($\text{PdCl}_2 \cdot 4\text{NH}_3$): Precipitates $2.87\ \text{g}$ of AgCl.
- Compound B ($\text{PtCl}_4 \cdot 2\text{HCl}$): Precipitates $0.00\ \text{g}$ of AgCl.
- Compound C ($\text{CoCl}_3 \cdot 4\text{NH}_3$): Precipitates $1.435\ \text{g}$ of AgCl.

Deduce the coordination formula, secondary valence of the metal, and the number of ions formed in solution for each compound. (Molar mass of $\text{AgCl} = 143.5\ \text{g mol}^{-1}$).

Solution Strategy:

1. Calculate the moles of AgCl precipitated per mole of the complex:
   $\text{Moles of AgCl} = \frac{\text{Mass of AgCl}}{143.5\ \text{g mol}^{-1}}$. Then,
   $\text{Moles of Cl}^- \text{ ionisable per mole of complex} = \frac{\text{Moles of AgCl}}{0.01\ \text{mol}}$.
2. Chloride ions outside the coordination brackets are ionisable; groups inside the brackets satisfy the secondary valence (coordination number).

Calculations:

- **Compound A:** Moles of $\text{AgCl} = \frac{2.87}{143.5} = 0.02\ \text{mol}$. Per mole of complex: $\frac{0.02}{0.01} = 2\ \text{mol Cl}^-$ ionisable.
  • *Formula:* $[\text{Pd}(\text{NH}_3)_4]\text{Cl}_2$.
  • *Secondary Valence of* Pd: 4.
  • *Total ions in solution:* 3 ($1\ [\text{Pd}(\text{NH}_3)_4]^{2+} + 2\ \text{Cl}^-$, 1:2 electrolyte).
- **Compound B:** Moles of $\text{AgCl} = 0.00\ \text{mol}$. No chloride ions reside outside the coordination sphere.
  • *Formula:* $\text{H}_2[\text{PtCl}_6]$.
  • *Secondary Valence of* Pt: 6.
  • *Total ions in solution:* 3 ($2\ \text{H}^+ + 1\ [\text{PtCl}_6]^{2-}$).
- **Compound C:** Moles of $\text{AgCl} = \frac{1.435}{143.5} = 0.01\ \text{mol}$. Per mole of complex: $\frac{0.01}{0.01} = 1\ \text{mol Cl}^-$ ionisable.
  • *Formula:* $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$.
  • *Secondary Valence of* Co: 6 (4 from NH_3 + 2 from Cl^-).
  • *Total ions in solution:* 2 ($1\ [\text{CoCl}_2(\text{NH}_3)_4]^+ + 1\ \text{Cl}^-$, 1:1 electrolyte).

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Worked Example 2: Deducing Formulas and IUPAC Names

Problem: Write systematic IUPAC names for:
(i) $[\mathrm{Pt(NH_3)_2Cl(NO_2)}]$
(ii) $\mathrm{K_3[Cr(C_2O_4)_3]}$
(iii) $[\mathrm{Co(NH_3)_5(CO_3)}]\mathrm{Cl}$
And formulate the chemical representation for *diamminechlorido(methanamine)platinum(II) chloride*.

Solution:

- **(i) $[\mathrm{Pt(NH_3)_2Cl(NO_2)}]$:** Neutral complex. Ligands: ammine (2), chlorido (1), nitrito-N (1). Alphabetical order: ammine < chlorido < nitrito-N. Metal oxidation state: $x + 2(0) + 1(-1) + 1(-1) = 0 \implies x = +2$.
  IUPAC Name: **Diamminechloridonitrito-N-platinum(II)**.
- **(ii) $\mathrm{K_3[Cr(C_2O_4)_3]}$:** Cation is potassium. Anionic complex has three oxalate ligands (oxalato). Chromium in anionic complex takes suffix '-ate' (chromate). Oxidation state: $3(+1) + x + 3(-2) = 0 \implies x = +3$.
  IUPAC Name: **Potassium trioxalatochromate(III)**.
- **(iii) $[\mathrm{Co(NH_3)_5(CO_3)}]\mathrm{Cl}$:** Cationic complex with ammine (5) and carbonato (1). Counter anion is chloride. Oxidation state of cobalt: $x + 5(0) + 1(-2) + 1(-1) = 0 \implies x = +3$.
  IUPAC Name: **Pentaamminecarbonatocobalt(III) chloride**.
- **Formula Formulation:** Central atom is Pt. Ligands: 2 ammine ($\mathrm{NH_3}$), 1 chlorido (Cl), 1 methanamine ($\mathrm{NH_2CH_3}$). Listing alphabetically: Cl, $\mathrm{NH_3}$, $\mathrm{NH_2CH_3}$. Oxidation state $+2$ with one $\mathrm{Cl^-}$ inside gives complex cation charge $+1$, requiring one $\mathrm{Cl^-}$ counter anion.
  Formula: $[\mathrm{PtCl(NH_3)_2(NH_2CH_3)}]\mathrm{Cl}$.

BOSONANCE

Worked Example 3: Stereoisomer Enumeration in Octahedral Complexes

Problem: Determine the total number of stereoisomers (geometrical + optical) and identify which forms are optically active for:
(a) $[\mathrm{CoCl_2(en)_2}]^+$
(b) $[\mathrm{Co(NH_3)_3Cl_3}]$

Solution:

- **(a) $[\mathrm{CoCl_2(en)_2}]^+$:**
  • *Geometrical isomers:* 2 (*cis* and *trans*).
  • *Optical activity of cis:* The *cis*-isomer has no plane of symmetry or center of inversion; it is **chiral** and exists as a non-superimposable pair of enantiomers (dextro d and laevo l). Both are optically active.
  • *Optical activity of trans:* The *trans*-isomer possesses a horizontal plane of symmetry passing through the metal and the two chloride ligands, as well as an inversion center. It is **achiral and optically inactive**.
  • *Total stereoisomers:* **3** (d-cis, l-cis, and trans).
- **(b) $[\mathrm{Co(NH_3)_3Cl_3}]$:**
  • Type $[\mathrm{Ma_3b_3}]$. Shows 2 geometrical isomers: **facial (fac)** and **meridional (mer)**.
  • Both the *fac* and *mer* isomers possess internal mirror planes of symmetry (σ_v). Neither is chiral.
  • *Total stereoisomers:* **2** (both optically inactive).

BOSONANCE

Worked Example 4: VBT Spin States & Magnetic Moments

Problem: The magnetic moment of $[\mathrm{MnCl_6}]^{3-}$ is $4.9\ \mathrm{BM}$, whereas that of $[\mathrm{Mn(CN)_6}]^{3-}$ is $2.8\ \mathrm{BM}$. Explain this experimental observation using Valence Bond Theory, specifying the hybridisation, geometry, and type of complex in each case. (Atomic number of $\mathrm{Mn} = 25$).

Solution:

1. Ground state electronic configuration of Mn ($Z = 25$): $[\mathrm{Ar}]\, 3d^5 4s^2$.
   In both complexes, manganese is in the $+3$ oxidation state: $\mathrm{Mn^{3+}} \rightarrow [\mathrm{Ar}]\, 3d^4 4s^0 4p^0 4d^0$.
2. For $[\mathrm{MnCl_6}]^{3-}$:
   • $\mu = 4.9\ \mathrm{BM} \implies \sqrt{n(n+2)} = 4.9 \implies n = 4$ unpaired electrons.
   • The weak-field ligand $\mathrm{Cl^-}$ cannot force electron pairing. The four $3d$ electrons remain unpaired in separate orbitals ($3d^4$).
   • To accommodate 6 chloride electron pairs, the outer $4s$, three $4p$, and two $4d$ orbitals undergo sp^3d^2 hybridisation.
   • Geometry: *Octahedral*; Type: **Outer orbital (high spin) complex**.
3. For $[\mathrm{Mn(CN)_6}]^{3-}$:
   • $\mu = 2.8\ \mathrm{BM} \implies \sqrt{n(n+2)} = 2.83 \implies n = 2$ unpaired electrons.
   • The strong-field ligand $\mathrm{CN^-}$ forces two of the four $3d$ electrons to pair up, leaving two unpaired electrons and vacating two inner $3d$ orbitals:

$$3d^4 \rightarrow [\uparrow\downarrow][\uparrow][\uparrow][\ \ ][\ \ ]$$

   • The two empty inner $3d$, one $4s$, and three $4p$ orbitals undergo d^2sp^3 hybridisation.
   • Geometry: *Octahedral*; Type: **Inner orbital (low spin) complex**.

BOSONANCE

Worked Example 5: CFSE Calculation & Electronic Configuration in Octahedral Field

Problem: A d^5 metal ion forms an octahedral complex with (i) a weak field ligand where $\Delta_o < P$, and (ii) a strong field ligand where $\Delta_o > P$. Write the ground-state d-orbital electron configuration, number of unpaired electrons, and calculate the CFSE in terms of Δ_o for each case.

Solution:

- **(i) Weak Field Ligand ($\Delta_o < P$, High Spin):**
  • Because pairing energy exceeds the splitting gap, electrons enter the e_g level before pairing:

$$\text{Configuration: } t_{2g}^3 e_g^2$$

  • *Unpaired electrons:* $n = 5$ (strongly paramagnetic, $\mu = 5.92\ \text{BM}$).
  • *CFSE Calculation:*

$$\text{CFSE} = [-0.4 \times n(t_{2g}) + 0.6 \times n(e_g)]\Delta_o = [-0.4(3) + 0.6(2)]\Delta_o = (-1.2 + 1.2)\Delta_o = \mathbf{0\ \Delta_o}$$

  Note: The CFSE of high-spin d^5 in an octahedral field is exactly zero!
- **(ii) Strong Field Ligand ($\Delta_o > P$, Low Spin):**
  • Because the splitting gap exceeds the pairing energy, all five electrons fill the lower t_{2g} level:

$$\text{Configuration: } t_{2g}^5 e_g^0$$

  • *Unpaired electrons:* $n = 1$ (weakly paramagnetic, $\mu = 1.73\ \text{BM}$).
  • *CFSE Calculation:*

$$\text{CFSE} = [-0.4(5) + 0.6(0)]\Delta_o + 2P = \mathbf{-2.0\ \Delta_o + 2P}$$

BOSONANCE

Worked Example 6: Synergic Bonding and Carbon–Oxygen Bond Order

Problem: Arrange the following isoelectronic transition metal hexacarbonyl complexes in order of increasing carbon–oxygen ($\mathrm{C-O}$) bond length:

$$[\mathrm{Mn(CO)_6}]^{+}, \quad [\mathrm{Cr(CO)_6}], \quad [\mathrm{V(CO)_6}]^{-}$$

Provide the scientific rationale based on the synergic bonding model.

Solution:

1. All three complexes are octahedral, homoleptic, and isoelectronic (d^6 transition metal centers).
2. In transition metal carbonyls, the $\mathrm{M \rightarrow CO}$ π-backbonding transfers electron density from filled metal d-orbitals into the antibonding π^* molecular orbitals of carbon monoxide.
3. As the negative charge on the complex increases (or positive charge decreases):

$$[\mathrm{Mn(CO)_6}]^{+} < [\mathrm{Cr(CO)_6}] < [\mathrm{V(CO)_6}]^{-}$$

   the electron density on the central metal ion increases markedly.
4. Greater electron density on $\mathrm{V^{-}}$ results in maximum π-back-donation into the π^* orbitals of CO, which lowers the $\mathrm{C-O}$ bond order to the greatest extent and causes the greatest lengthening of the $\mathrm{C-O}$ bond.
5. Conversely, $\mathrm{Mn^{+}}$ holds its d-electrons tightly, exhibiting the least π-back-donation; hence the $\mathrm{C-O}$ bond in $[\mathrm{Mn(CO)_6}]^{+}$ remains closest to a formal triple bond (shortest bond length).

Order of Increasing $\mathrm{C-O}$ Bond Length:

$$[\mathrm{Mn(CO)_6}]^{+} < [\mathrm{Cr(CO)_6}] < [\mathrm{V(CO)_6}]^{-}$$

Corresponding Order of $\mathrm{C-O}$ Bond Strength / Stretching Frequency (ν_{CO}):

$$[\mathrm{V(CO)_6}]^{-} < [\mathrm{Cr(CO)_6}] < [\mathrm{Mn(CO)_6}]^{+}$$

11. NEET UG Exam-Aligned Practice Questions

The following original practice questions simulate the conceptual depth, multi-statement structure, and rigorous format of the NEET UG Chemistry examination. Each question includes an unambiguous correct answer and detailed step-by-step reasoning.

BOSONANCE

Practice Question 1

Single-Correct MCQ • Werner's Theory

A coordination compound has the empirical formula $\mathrm{CoCl_3 \cdot 5NH_3}$. When $0.1\ \mathrm{mol}$ of this compound is treated with excess cold aqueous $\mathrm{AgNO_3}$, $28.7\ \mathrm{g}$ of AgCl is precipitated. The electrical conductivity of its aqueous solution corresponds to a:

(A) 1:1 electrolyte

(B) 1:2 electrolyte

(C) 1:3 electrolyte

(D) Non-electrolyte

Correct Answer: (B)

Worked Solution:

1. Calculate moles of AgCl precipitated:

$$\text{Moles of AgCl} = \frac{28.7\ \mathrm{g}}{143.5\ \mathrm{g\,mol^{-1}}} = 0.2\ \mathrm{mol}$$

2. Determine moles of ionisable $\mathrm{Cl^-}$ per mole of complex:

$$\frac{0.2\ \mathrm{mol\ AgCl}}{0.1\ \mathrm{mol\ complex}} = 2\ \mathrm{mol\ Cl^-\ per\ mol\ of\ complex}$$

3. Since 2 chloride ions are ionisable, they reside outside the coordination sphere. The remaining 1 chloride ion and all 5 $\mathrm{NH_3}$ molecules reside inside the sphere to satisfy the secondary valence (coordination number of $\mathrm{Co^{3+}} = 6$).

4. The modern formulation is $\mathrm{[CoCl(NH_3)_5]Cl_2}$. In aqueous solution, it dissociates as:

$$\mathrm{[CoCl(NH_3)_5]Cl_2 \rightarrow [CoCl(NH_3)_5]^{2+} + 2\,Cl^-}$$

producing 1 complex cation and 2 chloride anions, which corresponds to a **1:2 electrolyte**.

BOSONANCE

Practice Question 2

Single-Correct MCQ • IUPAC Nomenclature

According to current IUPAC conventions, the correct systematic name for the coordination compound $\mathbf{K_3[Fe(CN)_5(NO)] \cdot 2H_2O}$ containing neutral nitrosyl is:

(A) Potassium pentacyanidonitrosylferrate(II) dihydrate

(B) Potassium pentacyanidonitrosyliron(III) dihydrate

(C) Potassium nitrosylpentacyanidoferrate(III) dihydrate

(D) Tripotassium pentacyanonitrosylferrate(II) hydrate

Correct Answer: (A)

Worked Solution:
1. The cation is named first: *potassium*. (Never indicate the number of counter ions with prefixes like 'tripotassium').
2. Ligands inside the anionic coordination sphere are named alphabetically:
• 5 $\mathbf{CN^-} \rightarrow$ *pentacyanido* (starts with 'c')
• 1 $\mathbf{NO} \rightarrow$ *nitrosyl* (starts with 'n')
Alphabetical priority: *pentacyanido* precedes *nitrosyl*.
3. Central metal in an anionic complex takes the suffix **-ate** with its Latin stem: **ferrate**.
4. Oxidation state of iron: $3(+1) + x + 5(-1) + 1(0) = 0 \implies x - 2 = 0 \implies x = +2$, denoted as **(II)**.
5. Water of crystallization is appended as *dihydrate*.
Combining these yields: **Potassium pentacyanidonitrosylferrate(II) dihydrate**.

BOSONANCE

Practice Question 3

Single-Correct MCQ • Ligand Denticity & CN

The coordination number and oxidation state of the central metal in the complex $[\mathrm{Fe(en)_2(C_2O_4)}]\mathrm{NO_3}$ are, respectively:

(A) 6 and +2

(B) 4 and +3

(C) 6 and +3

(D) 5 and +2

Correct Answer: (C)

Worked Solution:

1. *Coordination Number (CN):*

• 'en' (ethane-1,2-diamine) is a didentate ligand: $2 \times 2 = 4$ donor atoms.

• $\mathrm{C_2O_4^{2-}}$ (oxalate) is also a didentate ligand: $1 \times 2 = 2$ donor atoms.

• Total $\mathrm{CN} = 4 + 2 = 6$.

2. *Oxidation State of* Fe*:*

• 'en' is neutral (charge 0); oxalate carries charge -2; nitrate counter ion carries charge -1.

• Let oxidation state of iron be x:

$$x + 2(0) + 1(-2) + 1(-1) = 0 \implies x - 3 = 0 \implies x = +3$$

Hence, coordination number is **6** and oxidation state is **+3**.

BOSONANCE

Practice Question 4

Single-Correct MCQ • Stereoisomerism & Chirality

Which one of the following coordination entities is chiral and will exhibit optical activity?

(A) *trans*-$[\mathrm{CrCl_2(ox)_2}]^{3-}$

(B) *trans*-$[\mathrm{CoCl_2(en)_2}]^{+}$

(C) *cis*-$[\mathrm{PtCl_2(NH_3)_2}]$

(D) *cis*-$[\mathrm{CrCl_2(ox)_2}]^{3-}$

Correct Answer: (D)

Worked Solution:

1. A molecule or complex ion is optically active (chiral) if and only if it lacks both a plane of symmetry (σ) and a center of inversion (i).
2. In *trans*-$[\mathrm{CrCl_2(ox)_2}]^{3-}$ and *trans*-$[\mathrm{CoCl_2(en)_2}]^{+}$, the two chloride ligands lie on opposite sides (180°), providing an equatorial plane of symmetry and an inversion center; both are achiral and optically inactive.
3. *cis*-$[\mathrm{PtCl_2(NH_3)_2}]$ is square planar. All square planar complexes possess a principal horizontal mirror plane coinciding with the molecular plane; therefore, square planar complexes are universally achiral and optically inactive.
4. In ***cis***-$[\mathrm{CrCl_2(ox)_2}]^{3-}$, the two bidentate oxalate chelate rings are positioned mutually perpendicular, and the two chloride ligands are adjacent (90°). It possesses neither a plane of symmetry nor a center of inversion. Its mirror image is non-superimposable; hence, it is **chiral and optically active**.

Practice Question 5

Single-Correct MCQ • Isomerism Types

The pair of compounds $[\mathrm{Cr(H_2O)_5Cl]Cl_2 \cdot H_2O}$ and $[\mathrm{Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O}$ represent which type of isomerism?

(A) Linkage isomerism

(B) Ionisation isomerism

(C) Solvate (hydrate) isomerism

(D) Coordination isomerism

Correct Answer: (C)

Worked Solution:

Compounds that possess identical overall composition but differ in whether water molecules are directly coordinated to the metal cation as ligands within the coordination sphere or present as uncoordinated solvent held in the crystal lattice are known as **solvate isomers** (or **hydrate isomers** when water is the solvent).

BOSONANCE

Practice Question 6

Single-Correct MCQ • VBT Magnetic Properties

Which one of the following species is diamagnetic and adopts a square planar geometry?

(A) $[\mathrm{NiCl_4}]^{2-}$

(B) $[\mathrm{Ni(CO)_4}]$

(C) $[\mathrm{Ni(CN)_4}]^{2-}$

(D) $[\mathrm{CuCl_4}]^{2-}$

Correct Answer: (C)

Worked Solution:
1. In $[\mathrm{Ni(CN)_4}]^{2-}$, nickel is in $+2$ oxidation state ($3d^8$). Cyanide ($\mathrm{CN^-}$) is a strong-field ligand that forces pairing of the two unpaired $3d$ electrons, vacating one $3d$ orbital. The hybridisation is dsp^2, resulting in a **square planar** geometry. Because all electrons are paired ($n = 0$), it is **diamagnetic**.
2. $[\mathrm{NiCl_4}]^{2-}$ has sp^3 hybridisation (tetrahedral) with 2 unpaired electrons (paramagnetic).
3. $[\mathrm{Ni(CO)_4}]$ is diamagnetic ($3d^{10}$), but has sp^3 hybridisation (tetrahedral).
4. $[\mathrm{CuCl_4}]^{2-}$ contains $\mathrm{Cu^{2+}}$ ($3d^9$), which possesses 1 unpaired electron and is paramagnetic.

Practice Question 7

Single-Correct MCQ • CFT Electron Configuration

For an octahedral complex with electronic configuration d^6, if $\Delta_o > P$, the crystal field configuration and spin-only magnetic moment are:

(A) $t_{2g}^4 e_g^2$ and $4.90\ \mathrm{BM}$

(B) $t_{2g}^6 e_g^0$ and $0.00\ \mathrm{BM}$

(C) $t_{2g}^5 e_g^1$ and $2.83\ \mathrm{BM}$

(D) $t_{2g}^6 e_g^0$ and $1.73\ \mathrm{BM}$

Correct Answer: (B)

Worked Solution:
1. When $\Delta_o > P$, the crystal field splitting energy is greater than the pairing energy (strong field condition).
2. It is energetically more favorable for electrons to pair up in the lower-energy t_{2g} orbitals rather than occupy the higher-energy e_g orbitals.
3. All six d-electrons pair up in the three t_{2g} orbitals, giving the ground-state configuration:

$$t_{2g}^6 e_g^0$$

4. Number of unpaired electrons $n = 0$. The complex is diamagnetic, and $\mu_s = \sqrt{0(2)} = 0.00\ \mathrm{BM}$.

BOSONANCE

Practice Question 8

Single-Correct MCQ • Tetrahedral Splitting

Which of the following statements regarding tetrahedral coordination entities is **incorrect**?

(A) The crystal field splitting is inverted relative to octahedral complexes.

(B) The splitting energy is related to octahedral splitting by $\Delta_t = \frac{4}{9}\Delta_o$.

(C) Low-spin configurations are readily and frequently observed in tetrahedral complexes.

(D) The 'g' subscript is not used for energy levels because tetrahedral complexes lack an inversion center.

Correct Answer: (C)

Worked Solution:
Statement (C) is false because $\Delta_t = \frac{4}{9}\Delta_o$. Since Δ_t is very small, it is virtually always smaller than the pairing energy ($\Delta_t < P$). The orbital separation is insufficient to force electron pairing; hence, **low-spin configurations are rarely, if ever, observed** in tetrahedral coordination entities. Statements (A), (B), and (D) are all scientifically accurate.

BOSONANCE

Practice Question 9

Single-Correct MCQ • Spectrochemical Series & Absorption

The correct order of increasing wavelength of absorption (λ_{max}) in the visible region for the complexes $[Ni(NO_2)_6]^{4-}$, $[Ni(NH_3)_6]^{2+}$, and $[Ni(H_2O)_6]^{2+}$ is:

(A) $[Ni(H_2O)_6]^{2+} < [Ni(NH_3)_6]^{2+} < [Ni(NO_2)_6]^{4-}$

(B) $[Ni(NO_2)_6]^{4-} < [Ni(NH_3)_6]^{2+} < [Ni(H_2O)_6]^{2+}$

(C) $[Ni(NH_3)_6]^{2+} < [Ni(H_2O)_6]^{2+} < [Ni(NO_2)_6]^{4-}$

(D) $[Ni(NO_2)_6]^{4-} < [Ni(H_2O)_6]^{2+} < [Ni(NH_3)_6]^{2+}$

Correct Answer: (B)

Worked Solution:

1. Absorption wavelength (λ) is inversely proportional to crystal field splitting energy (Δ_o):

$$\Delta_o = \frac{hc}{\lambda} \implies \lambda \propto \frac{1}{\Delta_o}$$

2. According to the spectrochemical series, ligand field strength increases in the order:

$$H_2O < NH_3 < NO_2^-$$

3. Therefore, the magnitude of crystal field splitting increases as:

$$\Delta_o([Ni(H_2O)_6]^{2+}) < \Delta_o([Ni(NH_3)_6]^{2+}) < \Delta_o([Ni(NO_2)_6]^{4-})$$

4. Inverting this order for wavelength (λ_{max}) gives:

$$[Ni(NO_2)_6]^{4-} < [Ni(NH_3)_6]^{2+} < [Ni(H_2O)_6]^{2+}$$

BOSONANCE

Practice Question 10

Single-Correct MCQ • Metal Carbonyl Synergic Bonding

In metal carbonyls, the $\mathrm{M-C}$ π-bond is formed by the donation of a pair of electrons from:

(A) The lone pair on carbonyl oxygen into a vacant metal d-orbital.

(B) The lone pair on carbonyl carbon into a vacant metal d-orbital.

(C) A filled metal d-orbital into the vacant antibonding π^* orbital of carbon monoxide.

(D) A filled metal d-orbital into the bonding π orbital of carbon monoxide.

Correct Answer: (C)

Worked Solution:
In transition metal carbonyls, the $\mathrm{M-C}$ σ-bond is formed by donation of the carbon lone pair into a vacant metal hybrid orbital. The $\mathrm{M-C}$ π-bond (back-donation) is formed by the donation of an electron pair from a **filled metal d-orbital into the empty antibonding π^* molecular orbital** of CO.

Practice Question 11

Assertion-Reason • Chelate Effect

Given below are two statements labeled as Assertion (A) and Reason (R):
Assertion (A): The complex $[\mathrm{Co(en)_3}]^{3+}$ is thermodynamically more stable than $[\mathrm{Co(NH_3)_6}]^{3+}$.
Reason (R): Ethane-1,2-diamine is a didentate chelating ligand that forms 5-membered rings with Co^{3+}, resulting in an entropy-driven chelate effect.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)

Worked Solution:
When a polydentate ligand binds to a metal ion, it forms ring structures called chelate rings. This chelate formation leads to an increase in entropy ($\Delta S^\circ > 0$) because fewer polydentate ligand molecules replace a greater number of unidentate ligands. Consequently, $[\mathrm{Co(en)_3}]^{3+}$ exhibits significantly higher thermodynamic stability than $[\mathrm{Co(NH_3)_6}]^{3+}$. Both (A) and (R) are true, and (R) correctly explains (A).

BOSONANCE

Practice Question 12

Statement-Based • Physical Color & Splitting

Examine the following two statements:
Statement I: Anhydrous $\mathrm{CuSO_4}$ is white, whereas hydrated $\mathrm{CuSO_4 \cdot 5H_2O}$ is blue in colour.
Statement II: In the absence of coordinating ligands, crystal field splitting does not occur, and consequently, $d - d$ electronic transitions cannot take place.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Answer: (A)

Worked Solution:
Statement I is correct: Anhydrous copper(II) sulphate has no coordinated ligands, making it white, whereas $\mathrm{CuSO_4 \cdot 5H_2O}$ contains $[\mathrm{Cu(H_2O)_4}]^{2+}$ chromophores that absorb red light and appear blue.
Statement II is also correct: Crystal field splitting requires the asymmetric electrostatic field of surrounding ligands. Without ligands, the five d-orbitals remain degenerate, preventing $d - d$ transitions. Thus, both statements are correct.

BOSONANCE

Practice Question 13

Match the Column • Applications of Coordination Compounds

Match the coordination compounds or reagents listed in **Column-I** with their primary scientific/medicinal applications in **Column-II**:

Column-I (Compound / Reagent)	Column-II (Application / Biological Role)
(a) Chlorophyll	(p) Homogeneous catalyst for hydrogenation of alkenes
(b) Cisplatin	(q) Chelation therapy for lead poisoning
(c) Wilkinson's catalyst	(r) Photosynthetic pigment containing magnesium
(d) Calcium disodium EDTA	(s) Antitumor chemotherapy drug

(A) (a) → (r), (b) → (s), (c) → (p), (d) → (q)

(B) (a) → (s), (b) → (r), (c) → (p), (d) → (q)

(C) (a) → (r), (b) → (p), (c) → (s), (d) → (q)

(D) (a) → (q), (b) → (s), (c) → (p), (d) → (r)

Correct Answer: (A)

Worked Solution:

• Chlorophyll is a coordination compound of magnesium responsible for photosynthesis: (a) → (r).
• Cisplatin (*cis*-$[PtCl_2(NH_3)_2]$) is an established antitumor chemotherapeutic agent: (b) → (s).
• Wilkinson's catalyst ($[(Ph_3P)_3RhCl]$) is used for homogeneous alkene hydrogenation: (c) → (p).
• Calcium disodium EDTA is used in chelation therapy to treat lead poisoning: (d) → (q).

BOSONANCE

Practice Question 14

Single-Correct MCQ • Metallurgy & Extraction

In the Mac-Arthur Forrest cyanide extraction process for gold, the water-soluble complex formed during aeration and the species used to displace metallic gold are, respectively:

(A) $[\mathrm{Au(CN)_4}]^-$ and Fe

(B) $[\mathrm{Au(CN)_2}]^-$ and Zn

(C) $[\mathrm{Au(CN)_4}]^{2-}$ and Cu

(D) $[\mathrm{Au(CN)_2}]^{2-}$ and Na

Correct Answer: (B)

Worked Solution:
Gold is oxidized by atmospheric oxygen in aqueous cyanide solution to form the soluble dicyanidoaurate(I) coordination entity:

$$4\,\mathrm{Au} + 8\,\mathrm{CN^-} + 2\,\mathrm{H_2O} + \mathrm{O_2} \rightarrow 4\,[\mathrm{Au(CN)_2}]^- + 4\,\mathrm{OH^-}$$

Gold is subsequently precipitated in metallic form by displacement with zinc powder:

$$2\,[\mathrm{Au(CN)_2}]^- + \mathrm{Zn} \rightarrow [\mathrm{Zn(CN)_4}]^{2-} + 2\,\mathrm{Au}\downarrow$$

Practice Question 15

Single-Correct MCQ • Carbonyl C-O Bond Order

Among the homoleptic metal carbonyls $[\mathrm{Fe(CO)_5}]$, $[\mathrm{Cr(CO)_6}]$, and $[\mathrm{Ni(CO)_4}]$, the total number of metal–carbon σ-bonds in a molecule of iron pentacarbonyl is:

(A) 3

(B) 5

(C) 10

(D) 0

Correct Answer: (B)

Worked Solution:
In $[\mathrm{Fe(CO)_5}]$, iron forms five coordinate σ-bonds by accepting lone pairs from the carbon atom of each of the five carbonyl ligands into its dsp^3 hybrid orbitals, adopting a trigonal bipyramidal geometry. Therefore, there are exactly **5** metal–carbon σ-bonds.

BOSONANCE

12. Common NEET Pitfalls & High-Yield Revision Checklist

Frequent Traps & How to Avoid Them

- **Double Counting in Coordination Number:** Remember that coordination number equals the number of σ-bonds, *not* the number of ligands. Always multiply the number of ligands by their denticity (e.g., in $[\mathrm{Fe}(\mathrm{C_2O_4})_3]^{3-}$, $\mathrm{CN} = 3 \times 2 = 6$, not 3!).
- **IUPAC Multipliers for Anions:** In naming ionic coordination salts, never include numerical prefixes for counter ions (e.g., $\mathrm{K_3[Fe(CN)_6]}$ is *potassium hexacyanidoferrate(III)*, never *tripotassium*).
- **Ambidentate vs. Chelating:** Ambidentate ligands (e.g., $\mathrm{NO_2^-}$, $\mathrm{SCN^-}$) have two donor atoms but only coordinate through *one* at a time to a single metal. They do not form chelate rings; they produce **linkage isomers**.
- **Optical Activity of Square Planar Complexes:** Square planar complexes are planar and always possess a molecular plane of symmetry; therefore, they *never* exhibit optical activity.
- **Tetrahedral Isomerism:** Tetrahedral complexes can never exhibit geometrical isomerism because all four positions are mutually adjacent (109.5°).
- **trans-Isomers with Chelates:** For $[\mathrm{M(AA)_2B_2}]$ (e.g., $[\mathrm{CoCl_2(en)_2}]^+$), only the **cis-isomer** is chiral and optically active. The **trans-isomer** has a plane of symmetry and is optically inactive.
- **Tetrahedral Spin States:** Never assign a low-spin configuration to a tetrahedral complex. Because $\Delta_\mathrm{t} = \frac{4}{9}\Delta_\mathrm{o} < \mathrm{P}$, tetrahedral complexes are always **high-spin**.
- **Zero CFSE for High-Spin d^5:** For $t_{2g}^3 e_g^2$, $\mathrm{CFSE} = [-0.4(3) + 0.6(2)]\Delta_\mathrm{o} = 0$.
- **Synergic Backbonding & C-O Bond:** Increased π-backbonding strengthens the $\mathrm{M-C}$ bond but *weakens* the $\mathrm{C-O}$ bond, increasing the $\mathrm{C-O}$ bond length and lowering its vibrational frequency (ν_{CO}).

NEET High-Yield Quick Revision Formula Sheet

- **Spin-Only Magnetic Moment:** $\mu_\mathrm{s} = \sqrt{n(n+2)}\ \mathrm{BM}$ ($n = \mathrm{unpaired\ electrons}$).
- **Tetrahedral vs. Octahedral Splitting:** $\Delta_\mathrm{t} = \frac{4}{9}\Delta_\mathrm{o}$.
- **Octahedral CFSE:** $\mathrm{CFSE_{oct}} = [-0.4\,n(t_{2g}) + 0.6\,n(e_g)]\Delta_\mathrm{o} + m\mathrm{P}$.
- **Tetrahedral CFSE:** $\mathrm{CFSE_{tet}} = [-0.6\,n(e) + 0.4\,n(t_2)]\Delta_\mathrm{t}$.
- **Spectrochemical Series:**
  $\mathrm{I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO}$
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13. Final Conceptual Summary

Coordination chemistry bridges fundamental inorganic bonding and modern materials/biochemical sciences. Alfred Werner established that transition metals satisfy both an ionisable primary valence (oxidation state) and a directional, non-ionisable secondary valence (coordination number). Ligands donate electron pairs into hybrid metal orbitals (VBT) or generate an electrostatic crystal field that lifts the degeneracy of the metal d-orbitals (CFT), splitting them into t_{2g} and e_g sets in octahedral fields, or

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inverted e and t_2 sets in tetrahedral fields ($\Delta_t = \frac{4}{9}\Delta_o$). The magnitude of splitting determines high-spin vs. low-spin electronic configurations, magnetic moments, and electronic absorption spectra responsible for the complementary colors of complexes and gemstones. In metal carbonyls, a unique synergic cycle of σ-donation and π-backbonding stabilizes zero-valent complexes. Across nature and industry, coordination compounds govern vital processes ranging from photosynthesis (chlorophyll) and oxygen transport (haemoglobin) to industrial alkene hydrogenation (Wilkinson's catalyst) and targeted cancer chemotherapy (cisplatin).

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